

Deep Learning-Based Vascular Region Estimation from Ultrasound Images

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Abstract—Vascular puncture requires operator’s skill, and even ultrasound-guided puncture relies on the operator’s ability to identify blood vessels from ultrasound images. This study aimed to develop a deep learning-based system for automatic vascular region estimation to support puncture procedures. Ultrasound images of the radial veins were acquired from eleven healthy adult males using a 10 MHz linear probe. A dataset comprising 9,305 annotated images was generated from the ultrasound videos, and a vascular segmentation model was developed using the architecture like U-Net. The model was trained for 200 epochs with a training-to-test ratio of 10:1 on a GPU-based workstation. The training and test accuracies stabilized after approximately 200 epochs. The model successfully detected the location and shape of blood vessels; however, the predicted vascular regions were slightly larger compared with the ground truth. These results demonstrate the feasibility of deep learning-based vascular localization from ultrasound images. Further optimization of the model and test using a larger cohort, including dialysis patients, are required for clinical application.

Keywords—vessel area, ultrasound image, deep neural network

I. INTRODUCTION

Vascular puncture is an invasive procedure, and its success largely depends on the operator’s skill. In addition, patient’s condition can substantially affect the success rate. In clinical practice, ultrasound-guided puncture has been proposed to improve the success rate of punctures. However, identifying blood vessels from ultrasound images still requires considerable expertise. This challenge is particularly important for patients undergoing home dialysis, who must perform self-puncture and may have impaired vision due to aging or diabetic retinopathy. Therefore, this study aims to develop a deep learning-based system for estimating blood vessel locations from ultrasound images and evaluate its accuracy.

II. METHODS

Eleven healthy adult males participated in this study. Radial veins in both arms were imaged in the short-axis view using a 10 MHz linear ultrasound probe. The acquired ultrasound videos were converted into images at 12 frames per second, and vascular regions identifiable within the images were manually annotated to generate a dataset of 9,305 images.

To prevent data leakage caused by including images from the same subject in different datasets, cross-test was performed on a subject-wise basis. Thus, all images obtained from the same subject, including consecutive frames from the same ultrasound video, were assigned exclusively to either the training or test set within each fold. The dataset was divided at the subject level with a training-to-test ratio of 10:1.

Data augmentation, including rotation and scaling, was applied to the training images to increase the diversity of the

training data. No data augmentation was applied to the test images. The vascular region estimation model was constructed using an architecture similar to U-Net and trained for 200 epochs.

Model performance was evaluated using pixel-level accuracy, F1-score, and intersection over union (IoU). Accuracy was calculated as the proportion of correctly classified pixels. The F1-score was calculated from the precision and recall of the vascular-region classification, and IoU was calculated as the intersection over union between the predicted vascular region and the ground-truth region. These metrics were monitored throughout training and evaluated on the test data in each cross-test fold.

Model training was performed using Windows 11 Pro (CPU: AMD Ryzen 5 4500, RAM: 64 GB, SSD: 500 GB, GPU: NVIDIA GeForce RTX 4070 SUPER 12 GB) and Python 3.11.

III. RESULTS

Figure 1 shows the changes in train and test loss during model training. The loss decreased rapidly during the early training period and subsequently decreased gradually. After approximately 100 epochs, the training loss remained relatively stable at a low level, while the test loss also showed a gradual decrease with fluctuations.

Figure 2 shows the changes in train and test accuracy. The accuracy increased rapidly during the early training period and reached approximately 0.95 by around 30 epochs. Thereafter, both training and test accuracy gradually increased and stabilized at approximately 0.98–0.99.

Figure 3 shows the changes in the F1-score during training. The F1-score gradually increased with the number of epochs and reached approximately 0.45 at the end of training. Although fluctuations were observed in the test results, the overall trend was similar to that of the training data.

Figure 4 shows the changes in the intersection over union

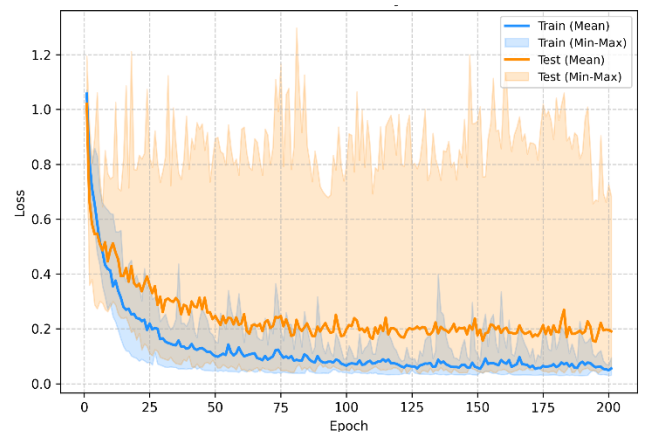


Fig. 1. Train and test loss during model training.

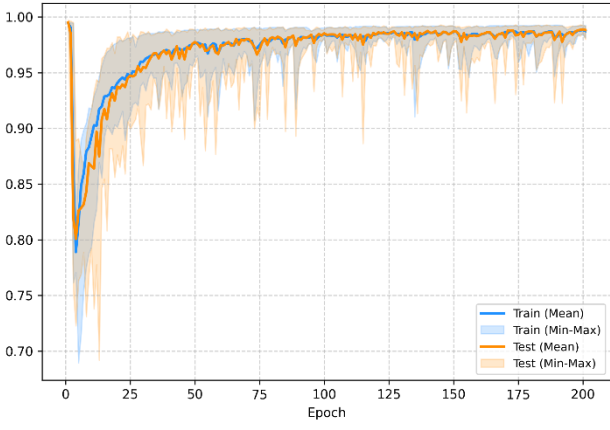


Fig. 2. Train and test accuracy during model training.

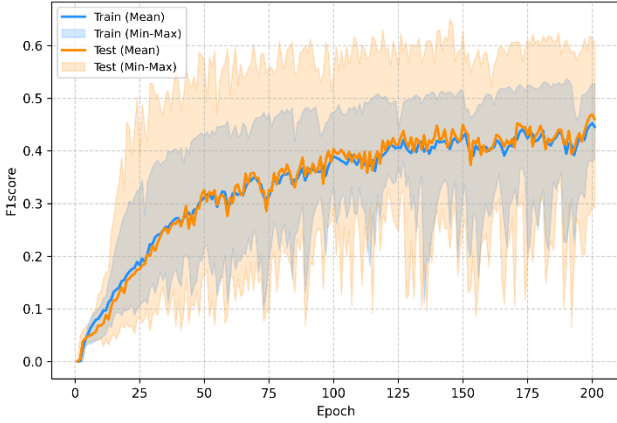


Fig. 3. Train and test F1-score during model training.

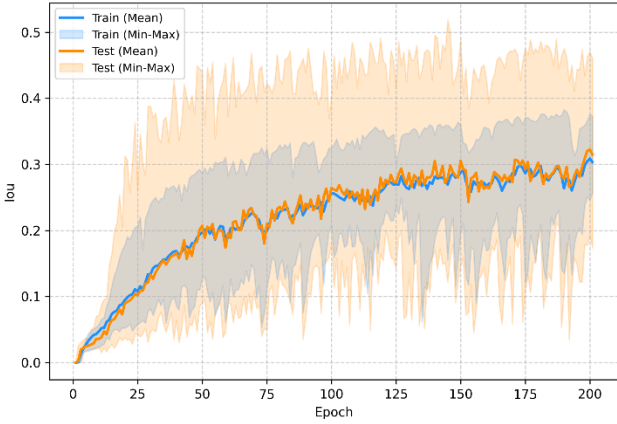


Fig. 4. Train and test IoU during model training.

(IoU). The IoU gradually increased during training and reached approximately 0.30 at the end of training. Similar to the F1-score, fluctuations were observed in the test results, but the overall performance improved with increasing training epochs. The accuracy was considerably higher than the F1-score and IoU. This difference suggests that pixel-level accuracy alone may not adequately represent the segmentation performance because the ultrasound images contain a relatively large background region. Therefore, F1-score and IoU were additionally used to evaluate the overlap between the predicted vascular regions and the ground-truth regions.

Fig. 5. Examples of vascular region estimation results. Representative ultrasound images, ground-truth vascular regions, and predicted vascular regions are shown for the

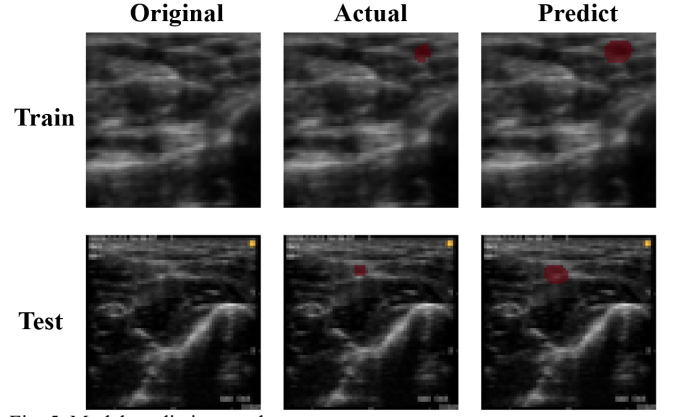


Fig. 5. Model prediction results

training and test datasets. Red regions indicate the annotated or predicted vascular regions.

Fig. 5. Examples of vascular region estimation results. Representative ultrasound images, ground-truth vascular regions, and predicted vascular regions are shown for the training and test datasets. Red regions indicate the annotated or predicted vascular regions.

IV. DISCUSSIONS

Previous studies have demonstrated the feasibility of deep learning-based blood vessel detection from ultrasound images [1]. However, studies targeting vessels in the forearm, which is an important site for vascular puncture, remain limited. In this study, a deep learning-based model was developed to estimate radial vein regions from ultrasound images, and its performance was evaluated using subject-wise cross-validation.

The accuracy increased rapidly during the early training phase and subsequently stabilized at approximately 0.98–0.99. However, the F1-score and IoU showed lower values, reaching approximately 0.45 and 0.30, respectively, at the end of training. This difference indicates that pixel-level accuracy alone may not adequately represent vascular segmentation performance. Because the ultrasound images contain a relatively large background area compared with the vascular regions, correct classification of background pixels can contribute substantially to the overall accuracy. Therefore, F1-score and IoU provide complementary information regarding the overlap between the predicted and ground-truth vascular regions. These results demonstrate the importance of using multiple evaluation metrics when assessing vascular region estimation from ultrasound images.

The prediction results shown in Fig. 5 indicate that the model was able to identify the general location of the radial veins in both the training and test datasets. However, the predicted vascular regions tended to be larger than the ground-truth regions. This tendency is consistent with the lower IoU and F1-score compared with the pixel-level accuracy. Precise identification of vascular boundaries can be difficult in ultrasound images because the boundary between the vessel and surrounding tissue is not always clearly defined. In addition, manual annotation of vascular regions may involve some degree of subjectivity. Further improvement of the segmentation model and annotation procedure may therefore be necessary to obtain more precise vascular boundaries.

An important feature of this study is that cross-validation was performed on a subject-wise basis. All images obtained from the same subject, including consecutive frames from the same ultrasound video, were assigned to the same fold. Thus, images from the same subject were not included in both the training and test datasets within a fold. This approach reduces the risk of data leakage caused by the high similarity between consecutive ultrasound images and provides a more appropriate evaluation of the model's generalization to previously unseen subjects.

The proposed method may have potential applications in ultrasound-guided vascular puncture. Automatic estimation of vascular regions could provide visual information to assist operators in identifying blood vessels from ultrasound images. Such assistance may be particularly useful in situations where identification of the vessel is difficult. However, the present study demonstrates the feasibility of automatic vascular region estimation rather than clinical effectiveness. The model was evaluated using ultrasound images from healthy adult males, and its ability to improve puncture accuracy or safety has not yet been established.

Several limitations should be considered. First, the dataset consisted of only eleven healthy adult males. The anatomical characteristics and ultrasound appearance of blood vessels may differ among individuals and patient populations. In particular, patients undergoing hemodialysis may show different vascular characteristics because of vascular access-related changes. Therefore, the generalizability of the proposed model to dialysis patients remains unclear. Second, this study focused on radial veins in the short-axis view. Further evaluation using long-axis ultrasound images and other vascular access sites is necessary. Third, although the model successfully estimated the general location of the vessels, the predicted regions tended to be larger than the ground-truth regions, indicating that further improvement in boundary accuracy is required.

Future studies should include a larger and more diverse population, including patients undergoing hemodialysis, and evaluate the model using both short- and long-axis ultrasound images. Further optimization of the segmentation architecture and training procedure may improve the accuracy of vascular boundary estimation. In addition, prospective evaluation during ultrasound-guided puncture procedures will be necessary to determine whether the proposed system can provide clinically meaningful assistance.

V. CONCLUSION

In this study, a deep learning-based model was developed to automatically estimate radial vein regions from ultrasound images. The model successfully identified the general location and shape of radial veins, demonstrating the feasibility of automatic vascular region estimation for ultrasound-guided puncture assistance. In addition to pixel-level accuracy, F1-score and IoU were used to evaluate segmentation performance, providing a more comprehensive assessment of the predicted vascular regions. Subject-wise cross-validation was also employed to reduce the risk of data leakage between training and test datasets.

Although the model was able to estimate vascular regions, the predicted regions tended to be larger than the ground-truth regions, indicating that further improvement in segmentation accuracy is required. Future studies should evaluate the proposed method using larger and more diverse datasets, including dialysis patients, and investigate its applicability to actual ultrasound-guided vascular puncture procedures.

REFERENCES

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